

In [1]:

```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.neighbors import KNeighborsClassifier
from sklearn.metrics import accuracy_score, classification_report
```

In [12]:

```
# Reading in data
```

```
data = pd.read_csv('D:/MIDS_PRAC/tweets.csv')
data.head()
```

Out[12]:

	tweet_id	airline_sentiment	airline_sentiment_confidence	negativereason	negativereason_
0	570306133677760513	neutral	1.0000	NaN	
1	570301130888122368	positive	0.3486	NaN	
2	570301083672813571	neutral	0.6837	NaN	
3	570301031407624196	negative	1.0000	Bad Flight	
4	570300817074462722	negative	1.0000	Can't Tell	

In [13]:

```
df = pd.DataFrame(data)
```

In [15]:

```
#split data
df.columns = df.columns.str.strip().str.lower()
print(df.columns)
```

```
Index(['tweet_id', 'airline_sentiment', 'airline_sentiment_confidence',
      'negativereason', 'negativereason_confidence', 'airline',
      'airline_sentiment_gold', 'name', 'negativereason_gold',
      'retweet_count', 'text', 'tweet_coord', 'tweet_created',
      'tweet_location', 'user_timezone'],
      dtype='object')
```

In [16]:

```
# Text vectorization using TF-IDF
vectorizer = TfidfVectorizer()
X_train_tfidf = vectorizer.fit_transform(X_train)
X_test_tfidf = vectorizer.transform(X_test)
```

In [17]:

```
# KNN Classifier
knn = KNeighborsClassifier(n_neighbors=3)
knn.fit(X_train_tfidf, y_train)
```

Out[17]:

▼ KNeighborsClassifier ⓘ ?

► Parameters

In [18]:

```
# Prediction
y_pred = knn.predict(X_test_tfidf)
```

In [20]:

```
# Evaluation
print("Accuracy:", accuracy_score(y_test, y_pred))
print("\nClassification Report:\n", classification_report(y_test, y_pred, zero_division=
```

Accuracy: 0.5

Classification Report:

	precision	recall	f1-score	support
0	1.00	0.50	0.67	2
1	0.00	0.00	0.00	0
accuracy			0.50	2
macro avg	0.50	0.25	0.33	2
weighted avg	1.00	0.50	0.67	2

In [21]:

```
print('''
Name: Ravindra kale
Class: TEAIML Batch: A2
Roll No:29
''')
```

Name: Ravindra kale
Class: TEAIML Batch: A2
Roll No:29

In []: